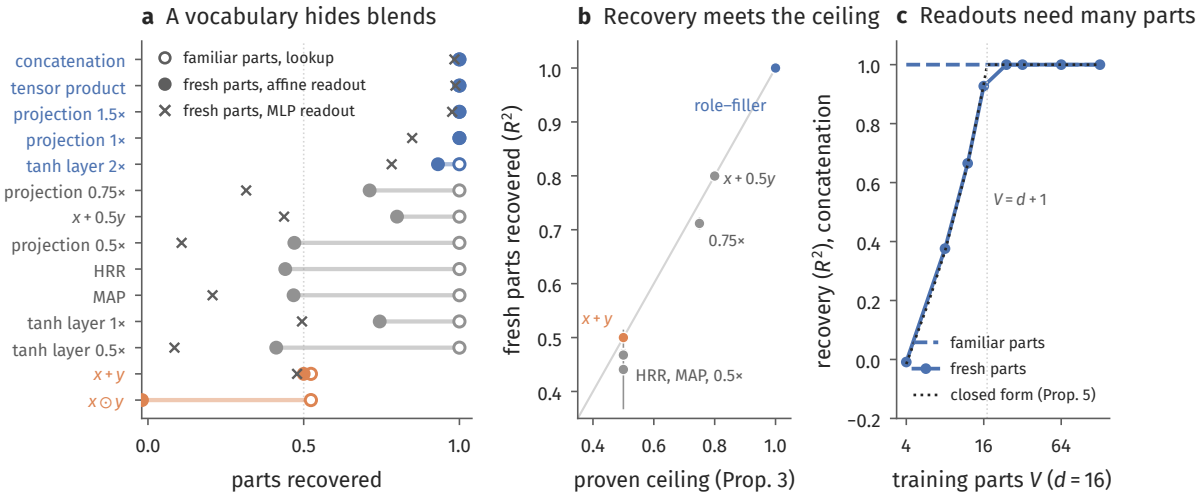

a A vocabulary hides blends
concatenation
tensor product
projection 1.5×
projection 1×
tanh layer 2×
projection 0.75×
x + 0.5y
projection 0.5×
HRR
MAP
tanh layer 1×
tanh layer 0.5×
x + y
x ⊙ y
familiar parts, lookup
fresh parts, affine readout
fresh parts, MLP readout
0.0
0.5
1.0
parts recovered
b Recovery meets the ceiling
fresh parts recovered (R^2)
role–filler
x + 0.5y
0.75×
x + y
HRR, MAP, 0.5×
0.4
0.6
0.8
1.0
proven ceiling (Prop. 3)
c Readouts need many parts
recovery (R^2), concatenation
V = d + 1
familiar parts
fresh parts
closed form (Prop. 5)
−0.2
0.0
0.2
0.4
0.6
0.8
1.0
4
16
64
training parts V (d = 16)